

Multiple Mean-shift Interval Detection for Multivariate Time Series with Applications to Stock Returns

Submitted for the Master of Science

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Summary

Change point detection is a problem of long-lasting interest in the field of statistics and time series analysis. Among different types of change-detection problems, epidemic change detection studies the scenario whereby the process departs abruptly from its baseline parametric description, but returns to it later. One prototypical sub-problem is epidemic mean change detection, whereby the mean of observed data deviates from its baseline and returns to it later, causing raised or dipped intervals.

In this dissertation, we propose a method for detecting multiple epidemic mean changes in multi-dimensional data series. The procedure is mostly based on taking aggregation of the scan statistics. We also included an adjustable sparsification bar to detect for signals with different density and filter out noises when taking aggregation. We assess the significance of detected mean changes by generating significance threshold on the maximum statistic using Rademacher multiplier bootstrapping.

We applied our methods to both simulated data and to real log-return data of stocks. In the simulation study, we demonstrated that methods based on the self-normalised scan statistics is more robust to heteroscedasticity than transitional non-self-normalised statistics. Under same settings of background noises, self-normalisation approach represents to have better accuracy in estimating the location of signals, as well as detecting the existence of a real signal. For the real data application part, we detected several significant trends by applying our methods on a multi-dimensional stock log return dataset.

1 Introduction

1.1 Background and Literature Review

A problem of long-standing interest in statistics and time series analysis is detecting whether and where the properties of the data sequence under observation change. Earliest study of this problem can be traced back at least to Page (1954). Changes can come in a variety of ways, e.g. (a) abrupt change in the parameters of the postulated statistical model, frequently referred to as change-point detection or break (point) detection; (b) change whereby the process departs abruptly from its baseline parametric description, but returns to it later, also known as epidemic change detection.

In this dissertation, we focus on problem of type (b) epidemic change detection. In particular, we are interested in investigating whether and where the average daily returns on exchange-traded stocks depart from its baseline behaviour, or in other words, whether and where significant trends in log-prices occur.

Study of problem type (b) can be traced back to Naus (1965a), Naus (1965b), while the term "epidemic change detection" is commonly used to refer to the this kind of problem after Worsley (1983) and Levin and Kline (1985), which studied the detection of rate changes in binomial, Poisson and retrospective binomial time series with application to epidemiology. Later, Siegmund (1986) studied boundary crossing probabilities to deal with epidemic detection problem, and Commenges et al. (1986) applied epidemic change-point model to investigating the relationship between neural discharges and stimulus, while Broemeling (1986) summarised several of its applications in econometrics.

One particular type of epidemic change detection is to detect mean-shift intervals. This problem with the baseline being zero mean has long been dealt with traditional scan statistics, see books Glaz et al. (1999), Glaz et al. (2001). With i.i.d. Gaussian noise setting and unknown baseline, Yao (1993) discussed several test statistics to detect such changes, including the likelihood ratio, the score-like statistic, Levin & Kline's statistic, the semi-likelihood ratio and the recursive residual. After Siegmund and Venkatraman (1995) who used the generalized likelihood ratio test (GLRT) to detect a single change point, Arias-Castro et al. (2005) used GLRT and multi-scale scan to detect elevated mean in data sequence and (as a regular shaped object) in space data, with usage in fields like image processing. After long time of evolution, Walther and Perry (2022) reviewed and compared existing scan statistics and proposed the self-normalised scan (SN Scan) statistic to deal with non-Gaussian, heteroscedastic observations with unknown baseline.

Epidemic detection for high-dimensional sequence is another problem of wide interest. Three important aspects about this problem is to: i) strengthen shared signals by combining information across dimensions; ii) eliminate the interference of added high-dimensional noises; iii) control the computational cost brought by high-dimensionality. There is a large amount of literatures who employed the idea of aggregation and sparsification to deal with such problem. We also refer to some important literatures of problem (a) that is related to our work, especially those uses cumulated sum (CUSUM) type statistics for high-dimensional time series. Early pioneering study on the method of aggregating CUSUM type statistics

cross-sectional include Aue et al. (2009), Zhang et al. (2010), Horváth and Hušková (2012) and Groen et al. (2013). Among which, Aue et al. (2009), Zhang et al. (2010) and Horváth and Hušková (2012) took ℓ_2 norm aggregation of CUSUM type statistic, Groen et al. (2013) investigated the maximum absolute and average CUSUM, which is the ℓ_∞ and ℓ_1 norm aggregation. Later, Cho and Fryzlewicz (2015) introduced the sparsified binary segmentation (SBS) algorithm which aggregates the CUSUM statistics by adding only those that pass a certain threshold when change points are sparse across the panel. Jirak (2015) further explored the ℓ_∞ norm to detect mean change in ARMA or GARCH series. Cho (2016) proposed the double-CUSUM statistic which has self-adaptiveness to different sparsity of signals. More recently, Yu and Chen (2020), Chen et al. (2022a) respectively studied finite sample and asymptotic performance of ℓ_∞ aggregated CUSUM and MOSUM statistic, while Li et al. (2024) proposed the ℓ_2 aggregation of MOSUM statistics, which favours nonsparse weak signals.

Detecting changes on average daily stock returns can also be modelled as epidemic mean change detecting problem. However, the background noise conditions for which can be fairly complex. Probable issues include temporal heteroscedasticity, cross-sectional heteroscedasticity and cross-sectional dependence for multi-dimensional financial time series. Since traditional scan statistic is sensitive to heteroscedasticity, it always requires estimating the standard deviation and doing pre-standardisation before testing. Those steps can be complex and may be inefficient if the wrong parametric assumption is used. Besides, traditional scan statistic is also less suitable for heavy tailed noise background because its level is largely influenced by extreme values. A scan statistic that is robust toward heteroscedasticity and heavy tails is proposed by Walther and Perry (2022), which has the form:

$$\mathrm{SN}_{[s,e]}(Y_t) = \frac{\sum_{t=s}^{e} Y_t}{\sqrt{\sum_{t=s}^{e} Y_t^2}},$$

where the only assumptions about Y_t is that they are independent and symmetrically distributed around zero, because according to Pinelis (2012), Bentkus and Dzindzalieta (2013) and Walther (2022), $\mathrm{SN}_{[s,e]}$ will satisfy a sub-Gaussian tail bound which is close to that of standard normal when Y_t satisfies pre-mentioned conditions. To the best of our knowledge, this SN Scan statistic has not been used in much change detection literatures, except for Fryzlewicz (2024) who proposed the self-normalised NSP method for univariate change point detection. In this dissertation, we propose to build its aggregated form and use it as the core statistic in our detection methods.

For completeness, we also refer to literatures of detecting shocks and trends in financial market. Two influential models based on this idea are the threshold autoregressive model of Tong (1983) and the Markov regime-switching (MRS) model of Hamilton (1989). Those methods are frequently employed by literatures to detect trends and shocks in stock market, see for examples Maheu and McCurdy (2000), Dendramis et al. (2014). Aiming for developing practical trading strategies, fast and real-time (online) change detection are very popular nowadays, see for examples Chen et al. (2022b), Halme et al. (2025), Kim et al. (2022). There are also vast literatures on trends detection using technical analysis methods and

machine learning algorithms, see Dragut (2012), Gudelek et al. (2017), Colliri and Zhao (2021). Many of the literatures in change point detection provided real data applications in detecting structural changes in high-dimensional stock returns, see Cho and Fryzlewicz (2015), Cho (2016), Jirak (2015), Li et al. (2024), but to the best of our knowledge, there was no study who used multi-dimensional epidemic mean change model to detect mean return changes and relate the elevated means to possible presences of trends.

In this study, we adapt several epidemic mean detection methods and compare their performances on simulated data and real daily stock return dataset. Methods included are: (a) conducting univariate detection using i) traditional scan statistic and ii) self-normalised scan statistics on each sequence and visualise the results to see possible clusters; (b) detect using aggregated scan statistics constructed by i) traditional scan statistic and ii) self-normalised scan statistics, each has three ways of taking aggregation, including ℓ_1 and ℓ_2 aggregation with an optional sparsifying step which respectively takes ℓ_1-norm and ℓ_2-norm of the scan statistics across dimensions, and the ℓ_∞ which takes the universal maximum of the scan statistic.

1.2 Outline

We first formulate our problem in Section 2.1, and state some notations and concepts we will use throughout the dissertation in 2.2. In Section 3, we construct two types of scan statistics which constitute the core of our methods and discuss some of their variants, as well as some important statistical properties, and then in Section 4, we formally bring up our main methodology. In Section 5, we examined the performance of our methods on different scenarios using simulated data. And finally in Section 7, we apply our methods in real stock analysis and present our finding in details, after introducing a small extension to our method which may be helpful in real data analysing in Section 6.

2 Problem Description and Notations

2.1 Problem Description

Consider the following problem: Let $X_1, \ldots, X_T$ be an observed p-dimensional time series satisfying

$$X_t = \beta + \mu(\frac{t}{T}) + \epsilon_t, \qquad t = 1, \ldots, T \tag{2.1}$$

where ϵ_t is a p-dimensional sequence of error terms symmetrically distributed around zero. That is, for $\epsilon_t = (\epsilon_t^{(1)}, \epsilon_t^{(2)}, \ldots, \epsilon_t^{(p)})^\top \in \mathbb{R}^p$, the expectations $\mathbb{E}(\epsilon_t^{(j)}) = 0$ and variances $\mathrm{Var}(\varepsilon_t^{(j)})$ unknown, $j = 1, \ldots, p$. We allow $\mathrm{Var}(\varepsilon_t^{(j)})$ to be different cross-sectionally and temporally. We combine the error terms into a matrix $Z = (\epsilon_1, \ldots, \epsilon_T) \in \mathbb{R}^{T \times p}$. In this study, we investigate the scenario with possible cross-sectional dependence but no serial

dependence. We formulate the cross-sectional dependence by the $p \times p$ covariance matrix:

$$\Sigma_t = \begin{bmatrix} \mathrm{Var}(\varepsilon_t^{(1)}) & \mathrm{Cov}(\varepsilon_t^{(1)}, \varepsilon_t^{(2)}) & \cdots & \mathrm{Cov}(\varepsilon_t^{(1)}, \varepsilon_t^{(p)}) \\ \mathrm{Cov}(\varepsilon_t^{(2)}, \varepsilon_t^{(1)}) & \mathrm{Var}(\varepsilon_t^{(2)}) & \cdots & \mathrm{Cov}(\varepsilon_t^{(2)}, \varepsilon_t^{(p)}) \\ \vdots & \vdots & \ddots & \vdots \\ \mathrm{Cov}(\varepsilon_t^{(p)}, \varepsilon_t^{(1)}) & \mathrm{Cov}(\varepsilon_t^{(p)}, \varepsilon_t^{(2)}) & \cdots & \mathrm{Var}(\varepsilon_t^{(p)}) \end{bmatrix} = \mathbb{E}[\varepsilon_t \varepsilon_t^\top],$$

and that

$$\mathbb{E}[\varepsilon_{t-k} \varepsilon_t^\top] = \mathbf{0}, \quad \forall k \neq 0.$$

$\mu(\cdot)$ is a p-dimensional vector of p unknown piecewise-constant functions denoted as $\mu^{(j)}$. Assume there are unknown $N^{(j)}$ non-adjacent mean-shift intervals in the process $\mu^{(j)}$ denoted by

$$[a_1^{(j)}, b_1^{(j)}], [a_2^{(j)}, b_2^{(j)}], \ldots, [a_{N^{(j)}}^{(j)}, b_{N^{(j)}}^{(j)}],$$

with

$$0 \leq a_1^{(j)} < b_1^{(j)} < a_2^{(j)} < b_2^{(j)} < \cdots < a_{N^{(j)}}^{(j)} < b_{N^{(j)}}^{(j)} \leq T$$

and

$$\mu^{(j)}\left(\frac{t}{T}\right) = \begin{cases} \mu_i^{(j)} \neq 0, t \in [a_i^{(j)}, b_i^{(j)}] \\ 0, t \notin [a_i^{(j)}, b_i^{(j)}] \end{cases}, \quad i = 1, 2, \ldots, N^{(j)},$$

where $\mu_i^{(j)} \neq 0$ are constants. Our goal is to detect those mean-shift intervals and find their start point and end point. The p-dimensional constant vector $\beta = (\beta^{(1)}, \beta^{(2)}, \ldots, \beta^{(p)})^\top \in \mathbb{R}^p$ represents the baseline. The baseline is possibly different for each dimensions, but is constant over time. The baseline can either be known or unknown, the later requires estimation from data. In this dissertation, we assumes a known zero-baseline for most of the study, while mention the unknown baseline version briefly in Section 3.

2.2 Notations and Definitions

In all following contents, we denote the observed time series as X_t, with the length of X_t denoted by T and the dimension as p. For multivariate time series, we denote as $X_t^{(j)}$ for the jth dimension of X_t. We denote by L the length of a signal, and denote by $[s_i^*, e_i^*]$ the true location of a signal, while denote the estimation of $[s_i^*, e_i^*]$ by $[\hat{s}_i^*, \hat{e}^*]$. Define ρ as the density of a common signal and m the number of dimension with that signal, ρ is the ratio of dimensions which has this signal over d. We denote the length of an interval $[s, e]$ by $n = e - s + 1$, and denote by $\mathcal{I}$ for a set of intervals $[s, e]$ satisfying $n \in [l, u]$, where l and u are the lower bound and upper bound of the length of signal. For a vector $v = (v_1, \ldots, v_d)^T \in \mathbb{R}^d$ and $q \in [1, \infty)$, we define its "ℓ_q-norms" "ℓ_∞-norm" and "ℓ_0-norm" respectively as

$$\|v\|_q = \left(\sum_{i=1}^{d} |v_i|^q\right)^{1/q} \quad , \quad \|v\|_\infty = \max_{1 \leq i \leq d} |v_i| \quad \text{and} \quad \|v\|_0 = \sum_{i=1}^{M} \mathbb{1}\{v_i \neq 0\}.$$

"ℓ_0-norm" here is not actually a norm, but uses the same form of notation, and write $a_n \asymp b_n$ to mean $0 < \liminf_{n\to\infty} \left|\frac{a_n}{b_n}\right| \leq \limsup_{n\to\infty} \left|\frac{a_n}{b_n}\right| < \infty$.

We also clarify some of frequently used concepts in this dissertation. We use the term "epidemic mean change" interchangeably with "mean-shift interval" and "signal". They all refer to the same type of change where the mean of the time series deviates from its baseline and return to it afterwards. Besides, note that we use multi-dimensional or multivariate, which is distinguished from high-dimensional even though the dimension is large, because in most literatures "high-dimensional" is defined by the asymptotic relationship of p and t, typically at least when $p_T \asymp T$ or sometimes $p_T \gg T$, see for example Enikeeva and Harchaoui (2013), Yu and Chen (2020). The density of signal is also often defined in asymptotic means. Under high-dimensional setting, signals are often defined as "dense" when $m_p \asymp p$, and weak when $\rho \to 0$ as $p \to \infty$. However in this dissertation, we investigate fixed T and p, and say that a signal is dense or sparse in a comparative sense, e.g. dense signal has a density ρ considerably larger then that of a sparse signal.

We use SN Scan short for self-normalised scan, and use non-SN Scan short for non-self-normalised scan. We use SN-ℓ_q and non-SN-ℓ_q to refer to method based on p-norm aggregation using SN or non-SN Scan statistic.

3 Constructing the Scan Statistics

The two forms of scan statistics, namely self-normalised scan statistic and traditional scan statistic, constitute the core statistics of the methodology proposed in this paper. Therefore, it is necessary to first investigate in some of their properties before bringing up the methodology.

For a univariate time series X_t, the traditional scan statistic, also known as the generalised likelihood ratio statistic, is constructed as:

$$\mathrm{S}_{[s,e]}(X_t) = \frac{\sum_{t=s}^{e} X_t}{\sqrt{e-s+1}}, \tag{3.1}$$

this statistic is widely used in scan problem under i.i.d. Gaussian noise background, see for references Glaz et al. (2001) and Arias-Castro et al. (2005). However, when the noise distribution is non-Gaussian, heavy-tailed or have greater variance, $\mathrm{S}_{[s,e]}$ may reach its peaks around extreme or high values caused by noises. To make this statistic robust for more complex variance setting, some literatures normalise this statistic by dividing a normalisation factor $\hat{\sigma}_t$, which is typically the standard deviation or the median absolute deviation (MAD) of X_t (see Jeng et al. (2012)). The former can be referred to as studentised local sum, see Csörgö and Horváth (1997), which is built as

$$\mathrm{S}_{[s,e]}(X_t) = \frac{\sum_{t=s}^{e} X_t}{\hat{\sigma}_t \cdot \sqrt{e-s+1}},$$

Consider possible existence of temporal heteroscedasticity, we pre-standardise the original data using a rolling window method instead for heteroscedastic or unknown variance cases:

$X'_t = X_t/\hat{\sigma}_t$, where

$$\hat{\sigma}_t = \frac{1}{2w}\sqrt{\sum_{i=t-w/2}^{t+w/2}(X_i - \frac{1}{2w+1}\sum_{i=t-w/2}^{t+w/2} X_i)^2},$$

the window-size is $2w+1$, and apply mirror padding to the data for cases where the rolling window extends beyond the data range.

To avoid local estimation of the deviance and in pursuit of robustness to non-Gaussian, heteroscedastic scenario, Walther and Perry (2022) proposed the self-normalised statistic is constructed of the form:

$$\mathrm{SN}_{[s,e]}(X_t) = \frac{\sum_{t=s}^{e} X_t}{\sqrt{\sum_{t=s}^{e} X_t^2}}, \tag{3.2}$$

it is scale-invariant because constant scaling factor applied to X_t will be cancelled out with its numerator and denominator. (3.2) can also be interpreted as $\sqrt{n}\cdot\cos\theta$, where θ is the angle of the vector $X_t[s,e] = (X_s, X_{s+1}, \ldots, X_{e-1}, X_e)$ and the unit vector $\mathbf{1}_n = (1,1,\ldots,1)$, and $n = e-s+1$ is the length of the interval. That is,

$$\sqrt{n}\cdot\cos\theta = \sqrt{n}\cdot\frac{x\cdot y}{\|x\|\,\|y\|} = \sqrt{n}\cdot\frac{Y_{[s,e]}\cdot\mathbf{1}_{e-s+1}}{\|Y_{[s,e]}\|\,\|\mathbf{1}_{e-s+1}\|} = \frac{\sum_{t=s}^{e} Y_t}{\sqrt{\sum_{t=s}^{e} Y_t^2}}. \tag{3.3}$$

The level of statistic (3.1) is therefore determined by the direction of the vector. The more it is align to the $\mathbf{1}_n$ vector, the larger the statistic would be.

More importantly, it is proved by Walther (2022) that $\mathrm{SN}_{[s,e]}(Y_t)$ satisfies a sub-Gaussian tail bound which is close to that of standard normal. For example, if Y_1 is a sample set of Cauchy distribution, Y_2 has a t-distribution with 5 degrees of freedom, and Y_3 has a double exponential distribution, then $\sum_{i=1}^{3} Y_i/\sqrt{\sum_{i=1}^{3} Y_i^2}$ will obey a guaranteed tail bound that is very close to the standard normal bound. To be specific,

$$\mathbb{P}\left(\frac{\sum_{t=s}^{e} Y_i}{\sqrt{\sum_{t=s}^{e} Y_i^2}} > a\right) \le \exp\left(-\frac{a^2}{2}\right).$$

Therefore, the self-normalised scan statistic can have better adaptiveness to heteroscedasticity than non-self-normalised scan. We will also provide empirical study results which can supports this argument in Section 5.

Since the scale of a trend is a crucial factor that should be considered in financial market, our proposed methods will have an additional step to bring the scale factor back after detection when it comes to applications, which will be described in Section 7.

For the case of dealing with unknown baseline, we briefly restate the method given in Walther and Perry (2022) here: set $n = e-s+1$, then divide the $T-n$ data outside the interval $[s,e]$ into n approximately equal partitions $J_t, t\in[s,e]$, with each sets no less than 2 elements, thus we have

$$\bigcup_{t\in[s,e]} J_t = [1,T]\setminus[s,e], \quad \bigcap_{t\in[s,e]} J_t = \emptyset, \quad \#J_t = \left\lfloor \frac{T}{n} - 1 \right\rfloor \text{ or } \left\lfloor \frac{T}{n} \right\rfloor \ge 2.$$

and then for $t \in [s, e]$, we calculate

$$\tilde{Y}_t = Y_t - \frac{1}{\#J_t} \sum_{j \in J_t} Y_j$$

and

$$\widetilde{\mathrm{SN}}_{[s,e]}(Y_t) = \frac{\sum_{t=s}^{e} \tilde{Y}_t}{\sqrt{\sum_{t=s}^{e} \tilde{Y}_t{}^2}},$$

$\tilde{\mathrm{S}}_{[s,e]}$ can be calculated similarly. This method incurs much higher computational cost than simply applying the known baseline. Since estimating the baseline is not the focus of this dissertation, unless otherwise stated, we assume the baseline to be zero throughout the remainder of this dissertation.

4 Methodology

4.1 Detection by Aggregation with Sparsification Options

4.1.1 Description of the Detection Procedure

Aggregation is a typically used technic for detecting common signals in multi-dimensional time series, especially those with a large number of components. Compared to conducting the detection process one by one on each single series, taking aggregation can greatly reduce computational costs and also avoid the problem of multiple testing. When signals are too weak to be detected in single series but are common and dense across dimensions, they can be cumulated by proper aggregation methods and therefore becomes detectable. However, aggregation also cumulates noises, which may drown some real signals, especially those are sparse. Adding a sparsification threshold and only aggregate those statistics that exceeded the threshold can help filtering out irrelevant noises and increase the detection power (seeCai et al. (2012), Cho and Fryzlewicz (2015)). In this dissertation, we use the following different ways of taking aggregation of scan statistics and adding an adjustable sparsification bar γ, which can also equals to 0, to aim for different types of signals:

i) (sparsified) ℓ_1 aggregation

$$\gamma \|v\|_{1,[s,e]} = \sum_{j=1}^{p} \left\{ \left| \mathrm{SN}_{[s,e]}^{(j)} \right| \cdot \mathbb{1}_{\left\{ \left| \mathrm{SN}_{[s,e]}^{(j)} \right| > \gamma \right\}} \right\};$$

ii) (sparsified) ℓ_2 aggregation

$$\gamma \|v\|_{2,[s,e]}^2 = \sum_{j=1}^{p} \left\{ \left(\mathrm{SN}_{[s,e]}^{(j)} \right)^2 \cdot \mathbb{1}_{\left\{ \left| \mathrm{SN}_{[s,e]}^{(j)} \right| > \gamma \right\}} \right\};$$

iii) ℓ_∞ aggregation:

$$\|v\|_{\infty,[s,e]} = \max_{1 \le j \le p} \left| \mathrm{SN}_{[s,e]}^{(j)} \right|;$$

Denoting by $\mathcal{I} = \{[s,e] : \ell_{\min} \leq e-s+1 \leq \ell_{\max}\}$ the collection of all intervals $[s,e]$ whose length lie between a predetermined upper and lower bound, we go through every possible interval $[s,e]$ to find the maximum of aggregated statistics:

$$\mathrm{U}^{\ell_\infty}(X_t) = \max_{[s,e]\in\mathcal{I}} \|v\|_{\infty,[s,e]} = \max_{[s,e]\in\mathcal{I}} \max_{1\leq j\leq p} \left|\mathrm{SN}^{(j)}_{[s,e]}\right|$$

$$\gamma\mathrm{U}^{\ell_1}(X_t) = \max_{[s,e]\in\mathcal{I}} \gamma\|v\|_{1,[s,e]} = \max_{[s,e]\in\mathcal{I}} \sum_{j=1}^{p} \left\{ \left|\mathrm{SN}^{(j)}_{[s,e]}\right| \cdot \mathbb{1}_{\left\{\left|\mathrm{SN}^{(j)}_{[s,e]}\right|>\gamma\right\}} \right\}$$

$$\gamma\mathrm{U}^{\ell_2}(X_t) = \max_{[s,e]\in\mathcal{I}} \gamma\|v\|^2_{2,[s,e]} = \max_{[s,e]\in\mathcal{I}} \sum_{j=1}^{p} \left\{ (\mathrm{SN}^{(j)}_{[s,e]})^2 \cdot \mathbb{1}_{\left\{\left|\mathrm{SN}^{(j)}_{[s,e]}\right|>\gamma\right\}} \right\}$$

and the corresponding intervals:

$$[\hat{s}^*, \hat{e}^*]^{\ell_1} = \arg\max_{[s,e]\in\mathcal{I}} \|v\|_{1,[s,e]},$$

$$[\hat{s}^*, \hat{e}^*]^{\ell_2} = \arg\max_{[s,e]\in\mathcal{I}} \|v\|^2_{2,[s,e]},$$

$$[\hat{s}^*, \hat{e}^*]^{\ell_\infty} = \arg\max_{[s,e]\in\mathcal{I}} \|v\|_{\infty,[s,e]}$$

as the estimation of the real signal region $[s^*, e^*]$.

We omit the counterpart for $\mathrm{S}_{[s,e]}$ here because it is exactly same so far, except the pre-standardization when the variance is unknown, which we discussed in Section 3.

To have a sense of how aggregation may work through using different norms q and setting different sparsification bar γ, we provide a little example here and visualise the procedure in Figure 1. sub-plot (a) shows the level and allocation of the mean-shifts. In a $T\times p = 500\times 500$ time series, there are three positive mean shifts: signal 1 start from $T = 100$ and end by $T = 150$, has a level of $\mu_1 = 0.2$ and is shared by 400 dimensions; signal 2 ranges from $T = 250$ to $T = 300$, has a level of $\mu_2 = 0.7$ and is shared by 20 dimensions; signal 3 ranges from $T = 350$ to $T = 500$, has a level of $\mu_3 = 0.4$ and is shared by 100 dimensions. The noise back ground for this example is i.i.d. t(5).

By setting the start point s to be the y-axis and the end point e to be the x-axis, we can visualise each matrix using a heat map, bright yellow represents where large aggregated statistics appear. According to Chan and Walther (2013), for mean-lift signal with unknown location and length in a single series, the lower bound of the power of signal $\mu\sqrt{L}/\sigma$ (L is the length of the signal) for the signal to be detected by traditional scan by probability 1 is $\sqrt{2\log T}$, if $\mu\sqrt{L}/\sigma \ll \sqrt{2\log T}$, then it is nearly impossible for even the optimal detection method to detect the existence of a signal. Signal 1 is almost undetectable by single series detection, however, as shared signals accumulates by ℓ_1 and ℓ_2 aggregation, its aggregated scan statistic now has a significantly high value (see the yellow spot in (c) and (d)). As we raise the sparsification bar, signal in becomes weaker while signal 3, which is the weakest in (c) and (d), becomes the strongest in (e) and (f). While is can also be note that More results about comparing each results will be shown in Section 5.

In the problem setting, there could be multiple real mean shifts regions, denote their loca-

tions $[s_1^*, e_1^*]$, $[s_2^*, e_2^*], \dots [s_N^*, e_N^*]$, where both N and the locations are unknown and requires inference. This can be formulate as a hypothesis test problem: for each given interval $[s, e]$, set H_0: $X_t,\ t \in [s, e]$ is pure noise; H_1: $[s, e]$ contains a true signal.

First of all, we conduct the following sequential detection procedure on the observed time series: search for the universal maximum ${}_\gamma\mathrm{U}^{\ell_q}(X_t)$ and compare it with the significance threshold C_α, if ${}_\gamma\mathrm{U}^{\ell_q}(X_t) \geq C_\alpha$, then we conclude that: at a significance level of α, $X_t,\ t \in [s_1^*, e_1^*]^{\ell_p}$ is not pure noise, i.e. it is a real mean shift. Then record ${}_\gamma\mathrm{U}^{\ell_q}(X_t)$ as ${}_\gamma\mathrm{U}_1^{\ell_q}(X_t)$, label the significance level of this test as $\alpha_1 = \alpha$.

After that, we remove all intervals that are joint with with $[\hat{s}_1^*, \hat{e}_1^*]^{\ell_p}$ from $\mathcal{I}$ and denote the new set of valid intervals as $\mathcal{I}_1$, search for ${}_\gamma\mathrm{U}^{\ell_q}(X_t)$ on new set $\mathcal{I}_1$, record it as ${}_\gamma\mathrm{U}_2^{\ell_p}(X_t)$ and the corresponding interval as $[s_2^*, e_2^*]^{\ell_p}$. If ${}_\gamma\mathrm{U}_2^{\ell_q}(X_t) \geq C_\alpha$, Label it as the second detected signal. We repeat this process until finding $\max_k\ {}_\gamma\mathrm{U}_k^{\ell_p}(X_t) \geq C_\alpha$, and record it as the kth signal.

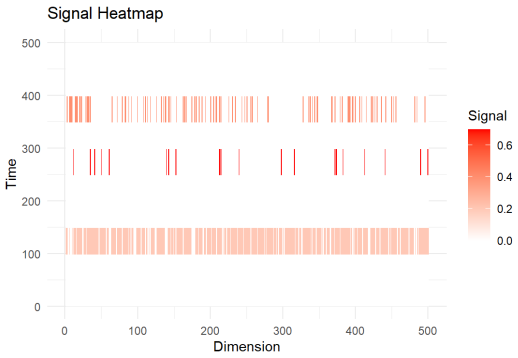
4.1.2 Estimating the Significance for Detected Intervals

How to decide the significance values for the detection procedure is a crucial problem. Most of literatures that focus on developing high-dimensional change detection algorithms adapt empirical critical values generated by bootstrapping. Chernozhukov et al. (2013) proposed the Gaussian multiplier (wild) bootstrap method to find the quantiles of the maximum of the sum of non-Gaussian high-dimensional random vectors. Later, Jirak (2015) introduced the Block Multiplier Bootstrap (BMB) and Düker et al. (2024) proposed the Cross-sectional & Temporal Block Wild (multiplier) Bootstrap (CT-BWB) to deal with high-dimensional mean change detection based on CUSUM statistics. Another multiplier that is often used is the Rademacher multiplier, which randomly changes the sign of samples. It can build simulations that are symmetrically distributed around zero even if the original data is asymmetric or biased, by randomly assigning multiplier ξ_t to centred X_t, where $\mathbb{P}(\xi_t = 1) = \mathbb{P}(\xi_t = -1) = \frac{1}{2}$. It is proposed first by Giné and Zinn (1990) and is proved to have strong control of the family-wise error rate with some adjustments by Arlot et al. (2010).

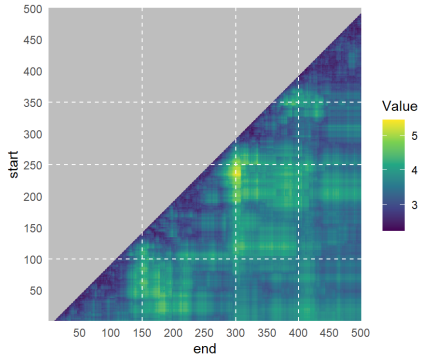
$$Y_t = \xi_t(X_t - \bar{X}_t), \quad \xi_t \overset{\text{i.i.d.}}{\sim} \text{Rademacher}, \tag{4.1}$$

Rademacher multiplier bootstrap is favourable because it is naturally align with our null hypothesis of white-noise of zero-baseline, while maintaining the original scale of observed data. It is able to preserve the cross-sectional dependence of the data, as well as the temporal localization of such dependence (i.e., the dependence structure stays concentrated in the same time regions).

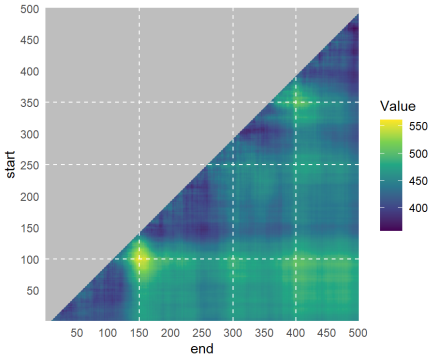
We generate the significance thresholds by the following procedure: do Rademacher multiplier bootstrap for B times for the observed time series to generate bootstrapped time series $Y_b,\ b \in [1, B]$. Searching for ${}_\gamma\mathrm{U}_1^{\ell_p}(Y_b)$ for each Y_b, and calculate the $(1-\alpha)$ percentile of ${}_\gamma\mathrm{U}_j^{\ell_p}$, α takes value in $0.01, 0.05, 0.1$, use these percentiles as C_α for the test.



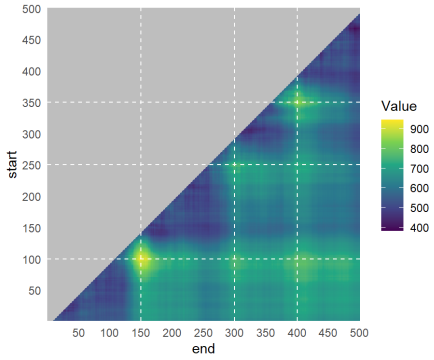
(a) allocation and Level of real signals



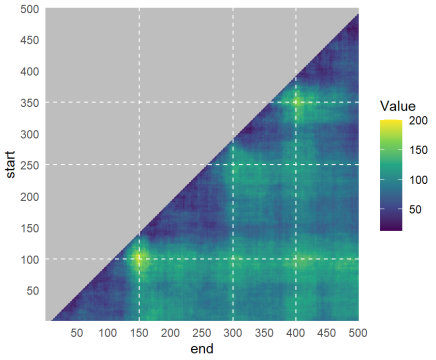
(b) SN-ℓ_∞ matrix



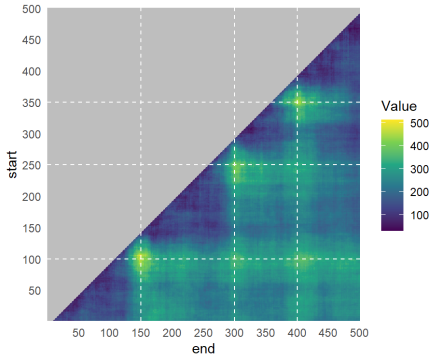
(c) SN-ℓ_1 matrix with $\gamma = 0$



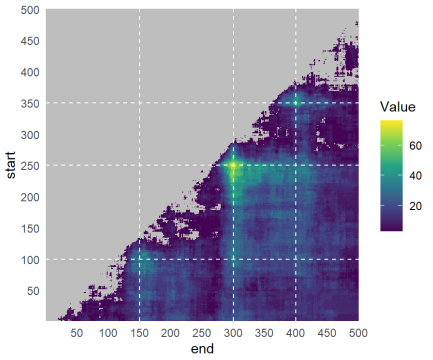
(d) SN-ℓ_2 matrix with $\gamma = 0$



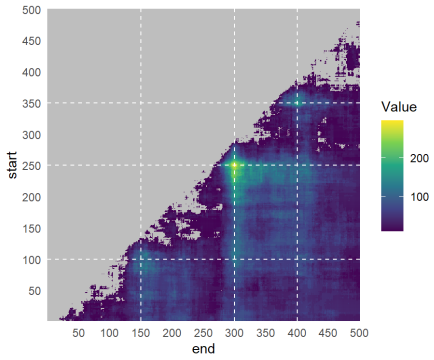
(e) SN-ℓ_1 matrix with $\gamma = 2$



(f) SN-ℓ_2 matrix with $\gamma = 2$



(g) SN-ℓ_1 matrix with $\gamma = 3$



(h) SN-ℓ_2 matrix with $\gamma = 3$

Figure 1: Demonstration of Aggregation Procedure

5 Simulation Study

In this section, we apply each methodology on randomly simulated data to analyse their performance under various noise structures and signal generation schemes, especially under heteroscedastic and cross-sectional correlation. For following simulations, unless otherwise stated, we control the dimension of simulated time series X_t as $p = 200$, the time length $T = 200$, which each real mean shifts (if any) located in the middle part of the time, i.e. $[s^*, e^*] \in [51, 150]$. Controlling for these basic variables allows us to eliminate their influence on the detection results, thereby getting a clearer view of the effects of heteroscedasticity and cross-sectional correlation.

We investigate in 3 types of signals:

(i) dense and weak signal;

(ii) semi-sparse medium signal;

(iii) strong and sparse signal

where we use the detective power to quantify the power of the assigned signals. The detective for single signal is $\text{PW}_{Single} = \mu\sqrt{L}/\sigma$. Note that defining the detective power is actually a non-trivial problem, in this dissertation we only need this concept in literature Chan and Walther (2013) to provide a rough assessment of the power of different types of signals. $\mu\sqrt{L}/\sigma$ The concept of detective power for high-dimensional epidemic mean change can be seen in Cai et al. (2012), we build high-dimensional signal detective power here as: $\text{PW}_{HD} = \frac{1}{\sqrt{d}} \sum_{j=1}^{p} \frac{\mu\sqrt{L}}{\sigma^{(j)}}$.

For noises, we consider the following scenarios:

(i) i.i.d. noise with known distribution;

(ii) cross-sectional heteroscedastic but temporally i.i.d. noise with known distribution;

(iii) temporally heteroscedastic noise process but being same for all dimensions.

Scenario i) aims to compare different methods in idealised settings, ii) aims to see if there is still any difference between the robustness of SN scan and standardized non-SN scan towards cross-sectional heteroscedasticity; while iii) aims to compare their robustness to temporal heteroscedasticity. Heteroscedasticity can have greater impacts to dense and weak signals compared to strong but sparse signals, because small signals are more easily to be drowned by high noises. Therefore, we build experiments about above three scenarios on weak and dense signal scenario, the density is $\rho = 0.75$, meaning that 150 dimensions over total 200 have a common signal. We set the signal magnitude to be constant across coordinates and time segments, so that heteroscedasticity naturally changes the effective SNR. The simulation scheme is designed as in Table 1.

Another factor to consider is cross-sectional correlation. Cross-sectional correlation can have large impact to dense and weak signals. It can make the noises raise or fall together, which will result in fake peaks after cumulated by ℓ_1 or ℓ_2 aggregation. It is hard to distinguish those peaks from real signals because they are actually very similar in nature. To be specific,

Table 1: Simulation design for different heteroscedasticity scenarios

Case	Noise distribution	sd	Signal level	SNR (single)	SNR (HD)
Case 1: homoscedasticity					
Case 1–1	i.i.d $N(0,1)$	1.00	0.10	0.55	5.81
Case 1–2	i.i.d $t(5)$	1.29	0.13	0.55	5.85
Case 1–3	i.i.d $t(5)$	1.29	0.10	0.42	4.50
Case 1–4	i.i.d $t(3)$	1.73	0.17	0.55	5.80
Case 1–5	i.i.d $t(3)$	1.73	0.13	0.41	4.36
1/3 of the dimensions are i.i.d $N(0,1)$*, 1/3 are i.i.d* $t(5)$*, and the rest are i.i.d* $t(3)$					
Case 2: cross-sectional heteroscedasticity					
Case 2		–	0.13	–	5.636
time blocked mix of $N(0,1)$ *and* $t(3)$*, block size=50*					
Case 3: temporal heteroscedasticity					
Case 3–0	pure noise	1.341	–	–	–
Case 3–1	signal in $N(0,1)$ region	1.341	0.10	0.41	4.33
Case 3–2	signal in $t(3)$ region	1.341	0.10	0.41	4.33
Case 3–3	signal across junction of two regions	1.341	0.10	0.41	4.33

*The standard deviation for Case 3 and for calculating its SNR is the population standard deviation of mixed noises.

if for a certain time region, random variables $X_t^{(1)}, X_t^{(2)} \dots X_t^{(d)}$ shares a constant correlation matrix with correlation coefficient equals ρ, i.e. $\Sigma = (1-\rho)I_d + \rho \mathbf{1}\mathbf{1}^\top$, $X_t^{(j)}$ can then be written by the form of a one-factor model as:

$$X_t^{(j)} = \sqrt{\rho} \cdot \sigma F_t + \sqrt{1-\rho} \cdot e_t^{(j)}, \tag{5.1}$$

where $e_t^{(j)}$ are independent with each other and with F_t, $\mathbb{E}[F_t] = \mathbb{E}[e_t^{(j)}] = 0$, $\mathrm{Var}(X) = \mathrm{Var}(e_t^{(j)}) = 1$. This did not bring real mean-shifts but can cause fake peaks when the sample size T is large. This can influence both detection and significance test in simulation study. To analyse the impact of correlation, we design the following experiments with signal level=0.13 and:

Table 2: Simulation design for correlated noise scenarios

Case	$t \in [201, 250]$	$t \in [301, 400]$	other time
Case 4–0	correlated noise	correlated noise	i.i.d. noise
Case 4–1	signal + correlated noise	i.i.d. noise	i.i.d. noise
Case 4–2	signal + i.i.d. noise	correlated noise	i.i.d. noise
Case 4–3	signal + correlated noise	correlated noise	i.i.d. noise

We use 3 key variables to evaluate the performance of adapted methods:

1. report rate: it estimates the proportion that the test statistic exceeds the 90% significance threshold over all simulations. When the tested series contains true signal, it represents the true positive rate. On the other hand, when the tested series is pure noise, it represents the false positive rate.

2. average IoU: IoU is short for Intersection over Union. For two intervals $[s, e]$ and

$[s^*, e^*]$,

$$\text{IoU}([s, e], [s^*, e^*]) = \frac{|[s, e] \cap [s^*, e^*]|}{|[s, e] \cup [s^*, e^*]|}.$$

We take the average IoU over all simulations to evaluate the accuracy of the estimation for the location of true signal.

3. success rate: For a simulated series containing a true signal, if the detection result is significant and the corresponding IoU ≥ 0.8, we define this as a "successful" detection. Success rate is defined as the proportion of "successful" detection over all simulations.

Testing results for each case is shown in Table 3, 4 and Table 5.

Table 3: Detection results for weak and dense signals against independent noises

Case	ℓ_1		ℓ_2		ℓ_∞	
	average IoU	IoU≥0.8	average IoU	IoU≥0.8	average IoU	IoU≥0.8
SN						
Case 1-1	0.872	0.85	0.912	0.95	0.134	0
Case 1-2	0.880	0.75	0.911	0.85	0.231	0.05
Case 1-3	0.351	0.05	0.601	0.35	0.157	0
Case 1-4	0.919	0.90	0.942	0.90	0.219	0.05
Case 1-5	0.746	0.55	0.829	0.70	0.130	0
Case 2	0.912	0.90	0.911	0.85	0.246	0
Case 3-0	0	0	0	0	0	0
Case 3-1	0.849	0.80	0.894	0.85	0.221	0.05
Case 3-2	0.337	0.15	0.489	0.35	0.135	0
Case 3-3	0.518	0.25	0.700	0.45	0.168	0
non-SN						
Case 1-1	0.889	0.90	0.892	0.80	0.064	0
Case 1-2	0.881	0.80	0.791	0.65	0.015	0
Case 1-3	0.442	0.20	0.405	0.15	0.0139	0
Case 1-4	0.887	0.80	0.191	0.15	0	0
Case 1-5	0.688	0.55	0.0333	0	0.008	0
Case 2	0.907	0.90	0.691	0.65	0.0184	0
non-SN with pre-testing standardisation (window=30)						
Case 3-0	0	0	0	0	0	0
Case 3-1	0.324	0.15	0.471	0.20	0.164	0
Case 3-2	0.409	0.30	0.561	0.45	0.147	0
Case 3-3	0.331	0.20	0.454	0.35	0.116	0
non-SN with pre-testing standardisation (window=75)						
Case 3-0	0	0	0	0	0	0
Case 3-1	0	0	0	0	0.00628	0
Case 3-2	0.693	0.60	0.502	0.40	0.160	0.05
Case 3-3	0.209	0	0.135	0	0.0566	0

Table 4: Case 3 Results part-2

Case	L1		L2		L∞	
	significance	Success Rate	significance	Success Rate	significance	Success Rate
			SN			
Case 3-0	0.1	0	0.133	0	0.133	0
Case 3-1	0.367	0.133	0.833	0.167	0.133	0
Case 3-2	0.233	0.033	0.2	0.033	0.1	0
Case 3-3	0.2	0	0.167	0.45	0.1	0
			non-SN, w=15			
Case 3-0	0.133	0	0.2	0	0.9	0
Case 3-1	0.267	0.15	0.233	0.033	0.933	0.033
Case 3-2	0.8	0.0667	0.3	0.0667	0.9	0.0667
Case 3-3	0.233	0	0.333	0	0.933	0
			non-SN, w=25			
Case 3-0	0.133	0	0.2	0	0.233	0
Case 3-1	0.133	0	0.2	0	0.267	0
Case 3-2	0.267	0.6	0.3	0.4	0.267	0.05
Case 3-3	0.133	0	0.233	0	0.233	0

Table 5: Case 4 Results-part 2

Case	L1		L2		L∞	
	significance	Success Rate	significance	Success Rate	significance	Success Rate
			SN			
Case 4-0	0.1	0	0.1	0	0.0333	0
Case 4-1	0.1	0.033	0.133	0.033	0.067	0
Case 4-2	0.133	0	0.1	0	0.1	0
Case 4-3	0.133	0	0.1	0	0.1	0
			non-SN, w=15			
Case 4-0	0.067	0	0.067	0	0.1	0
Case 4-1	0.1	0.033	0.067	0.033	0.067	0
Case 4-2	0.1	0	0.1	0	0.1	0
Case 4-3	0.067	0	0.067	0	0.1	0

When correlation exists inside the signal region, it can probably chop the signal into several pieces by converging to local peaks, with the first detected interval only returns a small piece of the whole signal. Cross-sectional correlation can also make the thresholds generated by the maximum scan statistic to be high so that real signals can hardly pass. However, although it is difficult to detect the real signal with one shot, the real signal is very likely to be covered in several detection results at the top. Different from the former, we use the sum of IoU of the top-k detected intervals to present the performance of the detection. The good news is that in stock return analysis, such second-order correlated segments are usually not undesirable since price co-movements also contain valuable information.

6 Extension: Detection by Single Series and Clustering Plots

Before going to real financial data application, we propose a small additional technique for epidemic mean change detection problem. In Section 4 we discussed the aggregation method which is suitable for detecting synchronous changes, however, asynchronous changes are also common in real-world application scenarios. For example, the impact of an external event

on individual stocks may be transmitted with heterogeneous time lags. As a complement to the main methodology, we introduce an innovative approach based on single series detection and visualisation, which may provide valuable implications for real data analysis.

Assume the input variable is a time series X_t of dimension p and length T, for each single series $X_t^{(j)}$, we search for the maximum of the scan statistic:

$$\mathrm{D}(X_t) = \max_{[s,e]\in[1,T]} \frac{|\sum_{t=s}^{e} X_t|}{\sqrt{\sum_{t=s}^{e} X_t^2}}, \tag{6.1}$$

or,

$$\mathrm{D}(X_t) = \max_{[s,e]\in[1,T]} \frac{|\sum_{t=s}^{e} X_t|}{\sqrt{e-s+1}}. \tag{6.2}$$

and the corresponding interval:

$$[\hat{s}^*, \hat{e}^*]^{(j)} = \arg \max_{[s,e]\in[1,T]} \frac{\left|\sum_{t=s}^{e} X_t^{(j)}\right|}{\sqrt{\sum_{t=s}^{e} (X_t^{(j)})^2}},$$

denote the result as $\mathrm{D}_1(X_t)^{(j)}$ and $[\hat{s}_1^*, \hat{e}_1^*]^{(j)}$. Similar to aggregation methods, we then search for the new $\mathrm{D}(X_t)^{(j)}$ and $[\hat{s}^*, \hat{e}^*]^{(j)}$ among intervals that are disjoint with $[\hat{s}_1^*, \hat{e}_1^*]^{(j)}$, denote the result as $\mathrm{D}_2(X_t)^{(j)}$ and $[\hat{s}_2^*, \hat{e}_2^*]^{(j)}$. This procedure can be continued to a certain depth to get k layers of results, representing k disjoint intervals with the largest k SN Scan statistics and their corresponding locations.

To further clarify the detecting procedure, we provide the following pseudocode:

Algorithm 1: Detection of epidemic mean shifts for each univariate series

Input : A multivariate time series X_t of size $T \times d$; number of layers k; the minimum interval length min_len; the maximum interval length max_len

Output: A table of $k \times d$ detected mean shifts: (label, layer, level, start, end, significance)

for $j \leftarrow 1$ **to** d **do**
  label $\leftarrow$ name of the jth sequence
  Initialize matrix $M \in \mathbb{R}^{T \times T}$ with all zeros
  for $s \leftarrow 1$ **to** $T - \text{max_len} + 1$ **do**
    for $e \leftarrow s + \text{min_len}$ **to** $s + \text{max_len} - 1$ **do**
      $M_{s,e} \leftarrow$ `Calculate_SN` $(X_t^{(j)}, s, e)$
    end
  end
  for $i \leftarrow 1$ **to** k **do**
    (start, end) $\leftarrow \arg\max_{s,e} |M_{s,e}|$
    level $\leftarrow M_{\text{start,end}}$
    layer $\leftarrow i$
    `// significance depends on specific methods of generating critical values`
    Record (label, layer, level, start, end, significance) in output table
    `// Mask overlapping regions in` M
    for $s \leftarrow 1$ **to** T **do**
      for $e \leftarrow s$ **to** T **do**
        if $e \geq$ start ***and*** $s \leq$ end **then**
          $M_{s,e} \leftarrow 0$
        end
      end
    end
  end
end

7 Detection on Real Data

7.1 Constructing the Dataset

For the real-data application, we demonstrate how our methods work on the multi-dimensional time series constructed by 5 years log-returns of NASDAQ-listed technology stocks. To ensure that all data is valid and clean, we only include stocks which has complete price data throughout 2020-7-20 to 2025-7-20 into our dataset, and use the adjusted prices to calculate log-returns. This dataset contains $p = 493$ dimensions and is of length $T = 1255$. Head rows are represented by Table 6.

Table 6: Demonstration of Components in the Dataset

Date	AAPL	MSFT	NVDA	⋯	MYSZ	FTFT	OBLG
2020/7/21	-0.013898	-0.013560	-0.017491		-0.033226	0.003945	-0.127513
2020/7/22	0.002805	0.014269	0.010618		-0.034368	-0.040166	0.381172
2020/7/23	-0.046585	-0.044469	-0.030048		-0.072496	-0.004107	0.055791
2020/7/24	-0.002480	-0.006141	0.006372		0.036905	0.067620	-0.010390
2020/7/27	0.023424	0.012588	0.022023		-0.029414	0.022815	-0.084404
2020/7/28	-0.016564	-0.009018	-0.019965		0.014815	-0.026668	-0.043548
			⋮				
2025/7/14	-0.012102	-0.000596	-0.005167		-0.063513	-0.415669	-0.074901
2025/7/15	0.002346	0.005551	0.039614		0.016261	0.007663	-0.024748
2025/7/16	0.005009	-0.000395	0.003917		0.039531	-0.152244	0.126101
2025/7/17	-0.000666	0.011953	0.009467		-0.007782	0.101352	0.021848
2025/7/18	0.005508	-0.003230	-0.003416		0.011651	0.043229	0.044195

7.2 Detection

We model it by a multiple epidemic mean shift model and try to detect possible mean shift intervals by our methods. An issue that we encountered at very beginning of the test is that, it was reported by the algorithm that no statistics, for any detection method, passed the significance threshold generated by the procedure in Section 4.1.2. We also tried simple permutation test, but there were no difference about the result.

For the searching procedure described in 4.1, we make a small modification here: instead of terminate the searching process as soon as the statistic failed to exceed the bound C_α, we directly search and record the largest K-layers of statistic, and observe how the statistic ${}_\gamma\mathrm{U}_k^{\ell_q}$ changes with the rank k. By setting $K = 20$, we plotted for example ${}_0\mathrm{U}_k^{\ell_2}(X_t), k = 1, \dots K$ in comparison of ${}_0\mathrm{U}_k^{\ell_2}(Y_{t,b})$ in Figure 2 and Figure 3. Remember that $Y_{t,b}, b = 1 \dots B$ are bootstrapped or permuted time series by X_t.

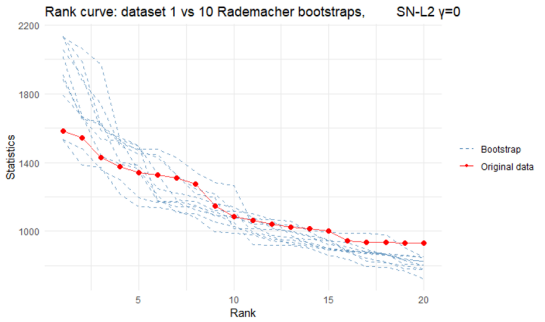


Figure 2: Rank curve compare with Rademacher bootstrap samples

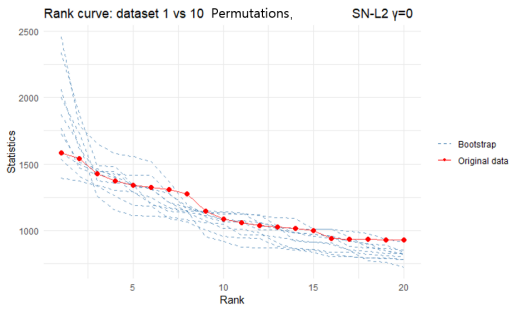


Figure 3: Rank curve compare with permutation samples

It is clear that our significance generating scheme lost its power in testing for this dataset. In light of conclusion of the simulation study, correlation structure can produce false peaks even when there is no true mean-changes, and boost the critical values. By applying the principle analysis (PCA) to dataset 1 (see Figure 4), it is shown that 25.54% of the variance

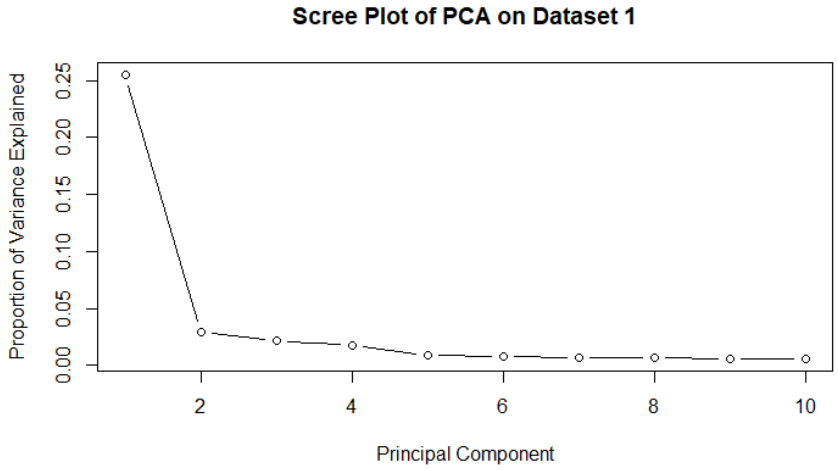


Figure 4: Variance explained by top 10 principle components

can be explained by the first principle component. In other words, their is a strong common market (or industry) factor influencing the log-returns of NASDAQ listed technology stocks.

Therefore, one possible reason that the test failed for real financial data is that the data presents strong correlation structure, while our method completely keeps the correlation structure when bootstrapping or permuting. This requires the signal to be much stronger to stand out than when there is only weak or no cross-sectional correlation.

However, our method can still uncover useful information from financial series, even without conducting a formal significance test, but by simply focusing on the ordering of the test statistics.

By applying both SN and non-SN methods with each aggregation methods and $\gamma = 0, 2, 3$ respectively, setting minimum interval length as 10 and maximum interval length as T to filter out those waves shorter than 2 weeks, we produced the result Table 7, which summarises the top 5 intervals returned by each method.

By combining common or similar intervals, we further refine the results to derive a list of worth noting intervals, see Table 8. The column methods (rank) labels the methods which return the interval at the rank. For example, the first period 2022-9-13 to 2022-9-26 is the interval with the largest statistic (rank 1) returned by SN-$\ell_1, \gamma = 0$, SN-$\ell_2, \gamma = 0$, SN-$\ell_1, \gamma = 2$ and SN-$\ell_2, \gamma = 2$.

Table 7: Detection results by $SN-\ell_q$ and non-$SN-\ell_q$ using different sparsification bars on Dataset 1

	SN			**non-SN** ($w = 15$ **for pre-standardisation**)		
Rank	statistic	start	end	statistic	start	end
ℓ_∞						
1	5.289	2022/8/18	2024/4/24	5.639	2021/8/9	2024/9/25
2	4.592	2020/10/8	2021/1/19	5.071	2020/10/8	2021/1/19
3	4.528	2024/5/22	2024/10/15	4.915	2024/9/27	2024/10/10
4	3.906	2021/8/2	2021/9/24	4.416	2024/10/24	2024/12/6
5	3.898	2025/4/14	2025/7/18	4.275	2025/4/30	2025/7/2
$\ell_1,\ \gamma = 0$						
1	829.829	2022/9/13	2022/9/26	800.688	2025/3/26	2025/4/8
2	807.172	2022/1/4	2022/1/27	797.475	2022/1/4	2022/1/27
3	768.188	2025/3/25	2025/4/8	789.089	2022/9/13	2022/9/26
4	759.576	2020/11/2	2021/2/16	772.639	2020/11/2	2021/2/16
5	733.639	2025/4/22	2025/5/13	747.996	2022/12/29	2023/2/2
$\ell_2,\ \gamma = 0$						
1	1584.986	2022/9/13	2022/9/26	1539.047	2022/1/4	2022/1/27
2	1543.409	2022/1/4	2022/1/27	1492.222	2020/11/2	2021/2/12
3	1428.934	2020/11/2	2021/2/16	1491.680	2022/9/13	2022/9/26
4	1376.333	2025/4/22	2025/5/13	1451.740	2025/3/26	2025/4/8
5	1340.508	2025/2/19	2025/4/8	1428.643	2023/10/31	2023/12/26
$\ell_1,\ \gamma = 2$						
1	385.289	2022/9/13	2022/9/26	374.060	2022/1/4	2022/1/27
2	369.949	2022/1/4	2022/1/27	371.290	2023/10/31	2023/12/26
3	343.037	2020/11/2	2021/2/12	362.910	2020/11/2	2021/2/12
4	335.946	2025/4/22	2025/5/13	330.689	2022/12/29	2023/2/2
5	332.209	2023/10/31	2023/12/26	322.578	2022/9/13	2022/9/26
$\ell_2,\ \gamma = 2$						
1	890.822	2022/9/13	2022/9/26	967.923	2023/10/31	2023/12/14
2	882.101	2022/1/4	2022/1/27	921.174	2020/11/2	2021/2/12
3	837.153	2020/11/2	2021/2/12	915.346	2022/1/4	2022/1/27
4	832.667	2023/10/30	2023/12/27	812.848	2022/12/29	2023/2/2
5	815.391	2025/4/22	2025/5/13	795.663	2022/9/13	2022/9/26
$\ell_1,\ \gamma = 3$						
1	65.508	2025/4/22	2025/7/3	93.197	2021/3/18	2025/4/8
2	54.638	2023/10/30	2023/12/27	61.216	2020/10/29	2021/2/16
3	42.366	2020/9/24	2021/1/14	54.367	2025/4/22	2025/7/3
4	38.413	2021/5/14	2021/9/3	26.625	2020/9/24	2020/10/9
5	28.808	2023/7/19	2023/10/13	12.873	2020/7/28	2020/8/10
$\ell_2,\ \gamma = 3$						
1	215.280	2025/4/22	2025/7/3	373.356	2021/3/18	2025/4/8
2	176.157	2023/10/30	2023/12/27	198.303	2020/10/29	2021/2/16
3	139.748	2020/9/24	2021/1/14	186.750	2025/4/22	2025/7/3
4	123.145	2021/5/14	2021/9/3	89.196	2020/9/24	2020/10/9
5	92.512	2023/7/19	2023/10/13	41.439	2020/7/28	2020/8/10

Table 8: Worth attention intervals and supporting methods

interval	start	end	methods (rank)
1	2022/09/13	2022/09/26	SN-$\ell_1, \gamma = 0$ (1); SN-$\ell_2, \gamma = 0$ (1); SN-$\ell_1, \gamma = 2$ (1); SN-$\ell_2, \gamma = 2$ (1); non-SN-$\ell_1, \gamma = 0$ (3); non-SN-$\ell_1, \gamma = 2$ (5); non-SN-$\ell_2, \gamma = 0$ (3); non-SN-$\ell_2, \gamma = 2$ (5)
2	2022/01/04	2022/01/27	SN-$\ell_1, \gamma = 0$ (2); SN-$\ell_2, \gamma = 0$ (2); SN-$\ell_1, \gamma = 2$ (2); SN-$\ell_2, \gamma = 2$ (2); non-SN-$\ell_1, \gamma = 0$ (2); non-SN-$\ell_2, \gamma = 0$ (1); non-SN-$\ell_2, \gamma = 2$ (3)
3	2020/11/02	2021/02/16	SN-$\ell_1, \gamma = 0$ (4); SN-$\ell_2, \gamma = 0$ (3); SN-$\ell_1, \gamma = 2$ (3); SN-$\ell_2, \gamma = 2$ (3); non-SN-$\ell_2, \gamma = 0$ (2); non-SN-$\ell_1, \gamma = 2$ (3)
4	2025/04/22	2025/05/31	SN-$\ell_1, \gamma = 0$ (5); SN-$\ell_2, \gamma = 0$ (4); SN-$\ell_1, \gamma = 2$ (4); SN-$\ell_2, \gamma = 2$ (5)
		2025/07/03	SN-$\ell_1, \gamma = 3$ (1); SN-$\ell_2, \gamma = 3$ (1); non-SN-$\ell_1, \gamma = 3$ (3); non-SN-$\ell_2, \gamma = 3$ (3); SN-ℓ_∞ (5); non-SN-ℓ_∞ (5)
5	2025/03/25	2025/04/08	SN-$\ell_1, \gamma = 0$ (3); non-SN-$\ell_1, \gamma = 0$ (1); non-SN-$\ell_2, \gamma = 0$ (4)
	2025/02/19		SN-$\ell_2, \gamma = 0$ (5)
6	2023/10/30	2023/12/27	SN-$\ell_2, \gamma = 2$ (4); SN-$\ell_1, \gamma = 3$ (2); SN-$\ell_2, \gamma = 3$ (2); non-SN-$\ell_1, \gamma = 2$ (2)
7	2020/09/24	2021/01/19	SN-$\ell_1, \gamma = 3$ (3); SN-$\ell_2, \gamma = 3$ (3); non-SN-$\ell_1, \gamma = 3$ (2); SN-ℓ_∞ (2); non-SN-ℓ_∞ (2)
8	2021/05/14	2021/09/03	SN-$\ell_1, \gamma = 3$ (4); SN-$\ell_2, \gamma = 3$ (4)
9	2022/12/29	2023/02/02	SN-$\ell_1, \gamma = 2$ (4); SN-$\ell_2, \gamma = 2$ (4); non-SN-$\ell_1, \gamma = 2$ (4)
10	2021/08/09	2024/09/25	non-SN-ℓ_∞ (1)
11	2022/08/18	2024/04/24	SN-ℓ_∞ (1)

7.3 Interpreting the Results

There is a large space for analysing these results. We propose several perspectives that may be of interest.

From the properties of different methods, we know that measures by SN is not sensitive to scale and may prefer stable trends, while non-SN is more sensitive to scale, and may prefer sharp trends. ℓ_1 and ℓ_2 with low sparsification bar prefer dense and common signals, while its preference for sparse signal increases as we raise γ, and ℓ_∞ prefers highly sparsified signals. Which means, intervals returned by lower γ are likely to contain common trends of the stocks, while with higher γ we may observe some movements shared by a smaller cluster of stocks. To manifest this, we first take a look at how the market or industry index behaves during those periods. For dataset 1, we use the he NASDAQ-100 Technology Sector Index (NDXT) as the industry index. NDXT includes 45 companies from the technology sector that are part of the NASDAQ-100 (up to 2025-7-20), and is often employed to capture the overall movements of the NASDAQ technology sector. We label those intervals on the price curve of NDXT, see Figure 5. Note that intervals returned only by SN-ℓ_∞ or non-SN-ℓ_∞ are not labelled.

Before looking into what is happening during those intervals in detail, we first propose several tools for analysis in this paragraph.

The first tool is "pruning", by which we can see which individual stocks contributed the

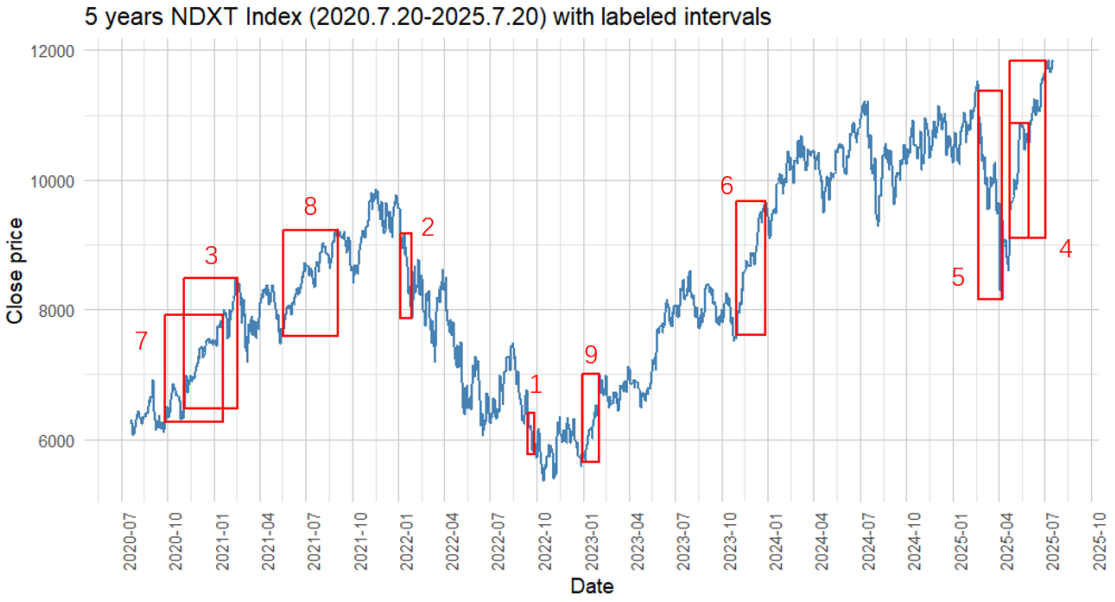


Figure 5: NDXT index labelled with intervals returned by aggregating methods

most to the aggregated statistics. For each reported interval $[s_i, e_i]$, we calculate dimension by dimension the univariate scan statistics in that interval. Then, we calculate an indicator named "contribution" for each stock:

$$\text{contribution}^{(j)}_{[s_i,e_i]} = \frac{S^{(j)}_{[s_i,e_i]}(X_t)}{\gamma \|v\|^2_{2,[s_i,e_i]}(X_t)} = \frac{\frac{\sum_{t=s}^{e} X_t^{(j)}}{\hat{\sigma}_t \cdot \sqrt{e-s+1}}}{\sum_{j=1}^{p} \left\{ \left| \text{SN}^{(j)}_{[s,e]} \right| \cdot \mathbb{1}_{\left\{ \left| \text{SN}^{(j)}_{[s,e]} \right| > \gamma \right\}} \right\}}.$$

After that, we rank all dimensions in descending order based on the magnitude of their contribution.

The second tool is to estimate the "level" of mean shifts (assume there is any). This is what we mentioned in Section 3 but was not discussed in detail. For each stock and time region $[s_i, e_i]$, we calculate the local mean as the estimation to the mean shift level:

$$\hat{\mu}^{(j)}_{[s,e]} = \bar{X}^{(j)}_{[s_i,e_i]} = \frac{1}{e_i - s_i + 1} \sum_{t \in [s_i,e_i]} X_t^{(j)}$$

Then we define the set of selected components $\mathcal{J}_{70}$ by taking the smallest set of indices along the reordered dimensions such that their cumulative contribution reaches (or just exceeds) 70%:

$$\mathcal{J}_{70} = \left\{ j : \frac{\sum_{d=1}^{j} \text{contribution}^{(i)}_{[s_i,e_i]}}{\sum_{m=1}^{p} \text{contribution}^{(m)}_{[s_i,e_i]}} \leq 0.7 \right\}.$$

Finally, the indicator "level" is defined as the weighted average of $\hat{\mu}^{(j)}_{[s,e]}$ over the selected

set $\mathcal{J}_{70}$, with weights given by the contributions:

$$\text{level}_{[s,e]} = \frac{\sum_{j\in\mathcal{J}_{70}} \text{contribution}^{(j)}_{[s_i,e_i]} \cdot \hat{\mu}^{(j)}_{[s,e]}}{\sum_{j\in\mathcal{J}_{70}} \text{contribution}^{(j)}_{[s_i,e_i]}}.$$

The last tool models the whole time series of an epidemic mean change model, and calculates the deviance from baseline explained by each detected mean-shifts. This is close to the idea of model-fitting, by which we can see the overall influence brought by deviations in each reported interval on the dataset. We name the indicator DEI in short of "deviance explained by each interval"

$$\text{DEI}_{[s_i,e_i]} = \frac{\sum_{j=1}^{p}\sum_{t=s_1}^{e_1}\left(X_t^{(j)} - \bar{X}^{(j)}_{[s_i,e_i]}\right)^2}{\sum_{j=1}^{p}\sum_{t=1}^{T}\left(X_t^{(j)}\right)^2}.$$

The DEI for above intervals 1-9 is plotted in 6. For intervals which have multiple start point or end point, we take the longest version of it.

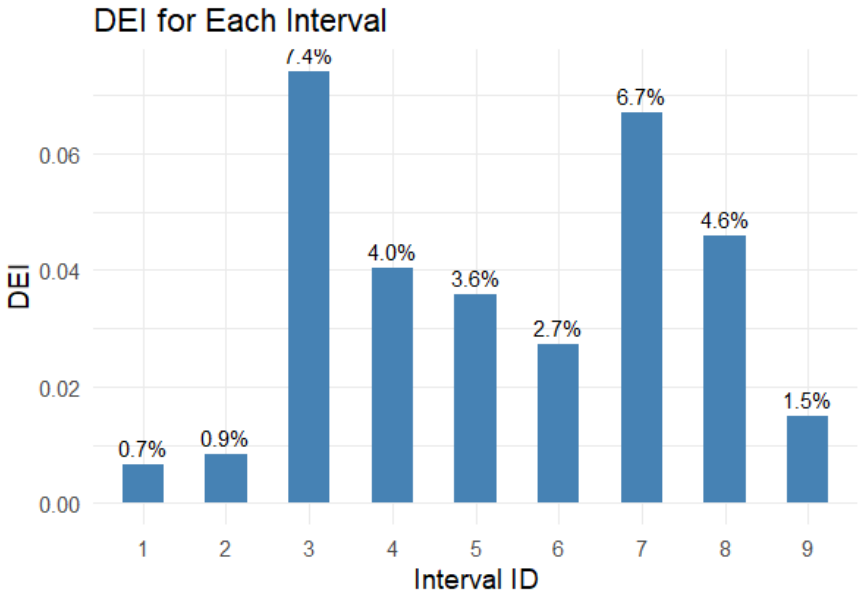


Figure 6: DEI of Intervals 1-9

Now we get back to Figure 5 and explain the findings in detail. Interval number 3, 4 and 5 are reported by low γ methods ($\gamma \leq 2$), where 5 is obviously ranked higher by non-SN methods. We can easily see significant downward or upward movement of the index in these intervals. By observing this, we can infer that NASDAQ technology stocks had common and obvious trends from November 2022 to February 2021, and in April and May of 2025. It is interesting to observe interval 7, which is about one month ahead of interval 3, is reported by several high γ approaches, which means the trend may be led by a smaller cluster of stocks, and followed by more stocks. Interval 3 and 7 also have the highest DEI (note that they are overlapped), which means this single trend itself explained over 7.4% of the deviation of log-return from 0. However, interval 1 and 2, which are ranked high by nearly all lower

γ approaches, looked trivial and short on the index. By investigating in the behaviours of individual stocks who have relatively large scan statistic in this interval, which means they are major contributors to the high aggregated statistic, we can observe that they had constant, nearly linear log-price decrease during that period. To see some individual price movements, we take SN-$\ell_1, \gamma = 0$ and interval 1 as example and find the stickers contributed the most to statistic, as presented by 9, and the plots of their log-prices is in 7.

Table 9: Top 9 stocks contribution to interval 1, by SN-$\ell_1, \gamma = 0$

Rank	Ticker	Statistic	Contribution
1	NTAP	-2.921066	0.003520081
2	AEVA	-2.818025	0.003395910
3	PAR	-2.809375	0.003385486
4	ERIC	-2.794605	0.003367688
5	FARO	-2.792370	0.003364994
6	GILT	-2.779679	0.003349700
7	BKSY	-2.742873	0.003305347
8	AMPG	-2.717301	0.003274206
9	LITE	-2.713441	0.003269879

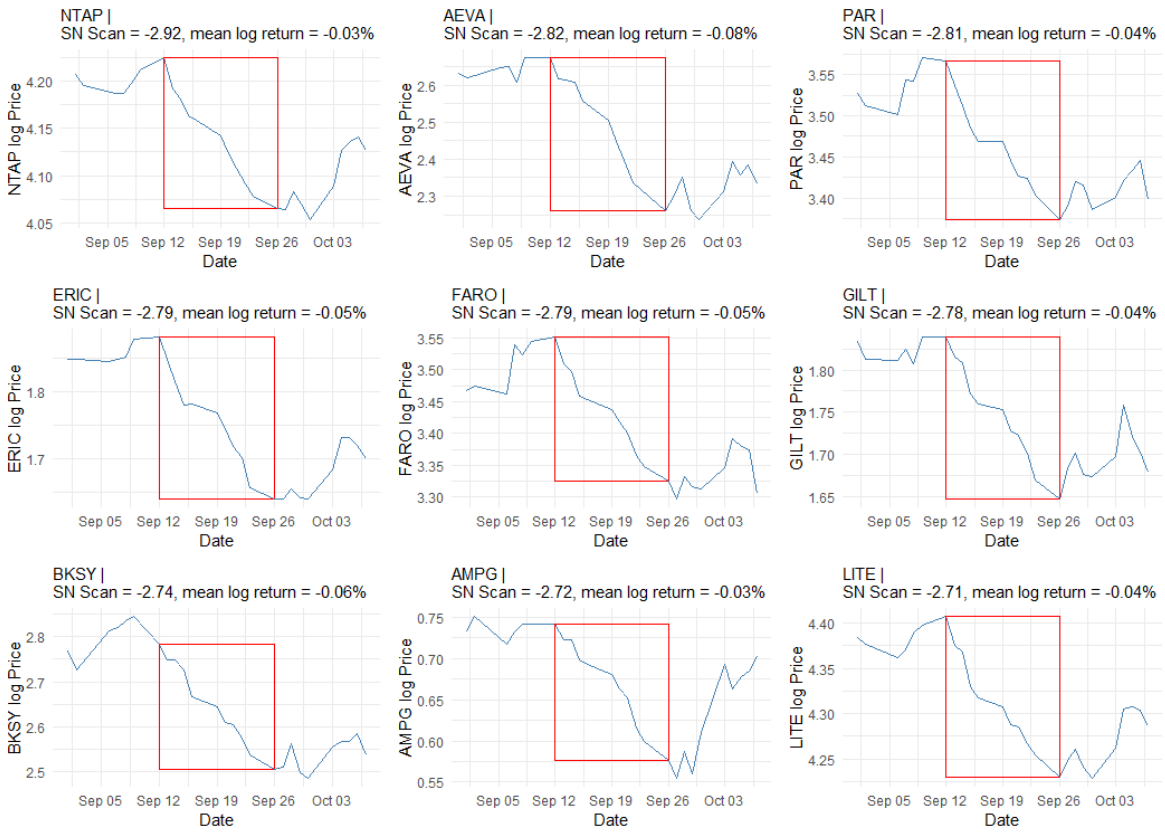


Figure 7: Log price shapes stocks contributed most to interval 1

Besides, log-returns in this interval also have very low variance, causing the denominator of SN scan being low and the pre-standardised data for non-SN scan being high. This could be the main reason why several of our statistics put this interval at the first place, higher than all trends that looks much more significant. But is this period really more trivial than

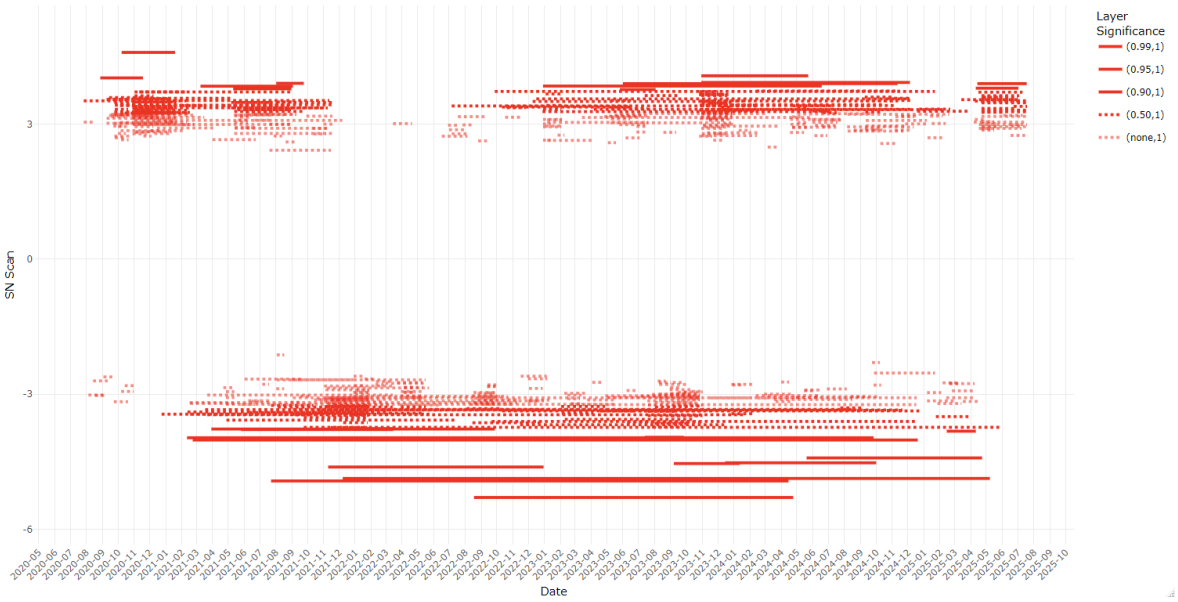


Figure 8: Cluster plot using SN scan

others as it seems like? In the next section, we provide some links of our observed results to real world events.

We finish this section by representing the "bar plot" using the method described in 6, see 8 and It is not difficult to spot from 8 that obvious positive clusters occurred in the following periods: November 2020 to January 2021, June 2021 to July 2021, November 2023 to December 2023, and April 2025 to May 2025; and negative clusters occurred in December 2021 to January 2022, and September to December, 2023.

7.4 A Linkage: Market Events that are Relevant to Our Results

In this extension, we investigate in whether some real world events happened in the intervals, which affected the log returns of NASDAQ technology stocks. One famous event would be the "Trump Slump" in April 2025, when President Trump announced sweeping tariffs, leading to dramatic and widespread panic-selling across markets, see Wikipedia contributors (2025). Interval 1 and 2 are all inside the bear market following the period of "everything bubble", see Wikipedia contributors (2024). In January 2022, the S&P 500, Nasdaq 100, and Dow 30 all dropped approximately 5.86%, 9.52%, and 3.97%. However, the trend was truncated the last two days of January, mainly because of impressive quarterly earnings of leading tech stocks, see Dhawan (2022) Following that, September 2020 was called "the worst September of Dow Jones in 20 years", see Saul (2022), when Dow Jones index fell over 5% during the month. The encouraging thing is that there are indeed much evidence to support that intervals detected by our methods are likely to be real shocks or trends.

8 Conclusion

In this dissertation, we proposed a robust procedure for epidemic mean change detection in multi-dimensional time series. With simulations under different scenarios, we found that

self-normalised scan statistic performs better than non-self-normalised scan statistic in heteroscedastic environment, representing an overall higher correctness of estimating the locations of signals. While in the scenario with existence of cross-sectional correlation, we found that the significance thresholds generated by Rademacher multiplier bootstrap could be over estimated, and the performance of the detection declined. To deal with this, we proposed to search for more intervals which has top ranking statistics rather than stop the detection procedure as soon as the top statistic fails to exceed the threshold. In real applications, it is suggested to investigate in all suspected signals and using further approaches to determine their significance. To do this, we proposed 3 indicators namely contribution, level and deviation explained by interval (DEI) for result analysis. By applying our methods to log return data of NASDAQ listed technology stocks, we observed 9 intervals that may contain real trends, and supported our detection results by referring to real world events.

9 Limitations and Other Extensions

One limitation is on the estimation for the real locations of signals. In this dissertation, we set the estimated location directly as the interval inside which the aggregated scan statistic reaches the maximum. There could be 2 directions to improving the estimation: firstly, it would be better to draw confidence intervals for the estimated start and end points rather than just doing point estimations; secondly, after identifying the interval with the maximum test statistic, we directly discard all overlapping intervals. This may cause a persistent signal to be fragmented or partially missed. A possible alternative could be to merge adjacent detected signals or to retain intervals with small overlap after each detection.

For another, results on the properties of methods and statistics are mostly empirical, also with some inference being intuitive, and lack of precise theoretical proof. For example the possible limiting distribution or asymptotic properties of aggregated SN scan, or any standardised procedure for choosing the sparsification bar, instead of just "guess".

From the application perspective, there can still be various extensions for utilise the aggregation scan procedure to analyse financial series. For instance, set the baseline β to be some market index and detect for deviations from the market, or whose returns outperform the market, or do one-direction aggregation on only positive or negative statistics instead of summing up absolute values, etc.

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Appendix 1: Acknowledgement: The use of generative AI tools

I acknowledge the use of GPT-4 and GPT-5 (https://chatgpt.com/) for assistance in programming, debugging, and LaTeX typesetting. The entire content of this paper was written independently by the author; however, some paragraphs benefited from translations provided by GPT from Chinese to English. During the literature review stage, GPT was also used to support reading, such as summarizing methodological contributions of papers and clarifying academic questions.